

Econ 306 Summer Session I

Quiz 1

You may not use any class materials or AI tools to answer the question. Write your answer on a separate piece of paper and then take a photo/scan your paper and upload to ELMS at the end of the quiz. You will have 20 minutes to complete the quiz and then 10 minutes to upload your answers. Round any decimal answers to 1 decimal place or leave your answers in fractional form (For example $\frac{50}{3}$). The quiz is out of 12 points. Please show your work as partial credit may be given for correct work. You are welcome to use a calculator.

A student at UMD on financial aid consumes Books (q_1) and Food (q_2). Her preferences are represented by the utility function:

$$U(q_1, q_2) = q_1^2 q_2$$

She has an initial budget of \$300 and the price of books is \$10 and the price of food is \$5.

1. Find the student's optimal consumption bundle (5 points).
2. Suppose the price of books rises to \$15 but income is unchanged. How much of each good do they purchase now? (5 points)?
3. How much income would she need to be able to afford her original consumption bundle (answer you found in Question 1) with the new price of books in Question 2 (2 points)?

Formula Sheet

Elasticities

$$E = \frac{dz}{dx} \frac{x}{z}$$

$$E_D = \frac{dQ_D}{dp} \frac{p}{Q_D}, E_S = \frac{dQ_S}{dp} \frac{p}{Q_S}, E_Y = \frac{dQ_D}{dY} \frac{Y}{Q_D}$$

MRS and MRT:

$$MRS = - \frac{MU_1}{MU_2}$$

$$\text{where } MU_i = \frac{\partial U}{\partial q_i}$$

$$MRT = - \frac{p_1}{p_2}$$

Perfect Substitutes

$$U(q_1, q_2) = aq_1 + bq_2$$

Perfect Complements

$$U(q_1, q_2) = \min \{aq_1, bq_2\}$$

Cobb Douglas

$$U(q_1, q_2) = q_1^a q_2^b$$

Cobb-Douglas MRS:

$$MRS = - \frac{aq_2}{bq_1}$$